

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: The correct answer is true for Statement 1 because every ideal is indeed a subring that is closed under addition and multiplication, but false for Statement 2 because not every subring satisfies the condition of absorbing multiplication from the ring, which is required to be an ideal.

2. **Answer:** D

Options: A) 0.64; B) 0.72; C) 0.81; D) 0.95

Note: The probability that neither integer chosen will be the square of the other is 0.95 because out of the 100 possible pairs of integers (10 choices for Sofia and 10 for Tess), only 5 pairs correspond to one integer being the square of the other (1 and 1, 2 and 4, 4 and 2, 3 and 9, 9 and 3), thus resulting in a probability of 95 out of 100 that they do not choose such pairs.

3. **Answer:** A

Options: A) $(x - 2)(x + 2)(x - 1)(x + 1)$; B) $(x+1)^4$; C) $(x-1)(x+1)^3$; D) $(x-1)^3(x+1)$

Note: The polynomial $x^4 + 4$ can be factored into linear factors in $\mathbb{Z}_5[x]$ as $(x - 2)(x + 2)(x - 1)(x + 1)$ because in the field $\mathbb{Z}_5$, the roots of the equation $x^4 + 4 = 0$ correspond to the values of x that produce zero when substituted into the polynomial, specifically at $x = 2, -2, 1$, and -1 , which simplifies to 2, 3, 1, and 4 modulo 5.

4. **Answer:** C

Options: A) $g = g^{-1}$ for every g in G ; B) $g = g^2$ for every g in G ; C) $(g \circ h)^2 = g^2 \circ h^2$ for every g, h in G ; D) G is of finite order

Note: In an abelian group, the operation is commutative, allowing the expression $(g \circ h)^2$ to be rewritten as $g \circ h \circ g \circ h$, which can be rearranged due to commutativity to equal $g \circ g \circ h \circ h$, confirming that $(g \circ h)^2 = g^2 \circ h^2$ for all elements g and h in the group G .

5. **Answer:** A

Options: A) Sunday; B) Monday; C) Tuesday; D) Wednesday

Note: Since the number of days can be reduced modulo 7 (the number of days in a week), calculating $10^{10^{10}} \mod 7$ yields 0, indicating that $10^{(10^{10})}$ days from Wednesday falls on the same day.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Every group of prime-power order is solvable because it can be shown that any group whose order is a power of a prime number has a normal subgroup whose index is a prime, leading to a series of normal subgroups that ultimately results in an abelian group.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the determinant of a 5×5 matrix can be non-zero even if some column vectors are linearly independent, as long as the entire set of columns is also linearly independent.

8. **Answer:** False

Options: A) True; B) False

Note: The equation $\cos(97x) = x$ has only one positive solution due to the behavior of the cosine function, which oscillates between -1 and 1, while the line $y = x$ continuously increases without bound, resulting in a maximum of one intersection point in the positive domain.

9. **Answer:** True

Options: A) True; B) False

Note: The dimension of the intersection of two 2-dimensional subspaces of R^4 can be 0, 1, or 2 because the intersection can vary from being empty (0-dimensional) to sharing an entire plane (2-dimensional), with 1-dimensional intersections representing cases where they share a line.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because, using the normal approximation to the binomial distribution, the probability of rolling a six 70 or more times in 360 tosses of a fair die falls within the calculated range of approximately 0.02 to 0.16.

Long Form Response

11. **Answer:** Let a be the number of colonies on Earth-like planets and b be the number on Mars-like planets. We therefore seek nonnegative integers a and b such that $2a + b = 12$. From this equation, we see that b can be at most 6 while a can at most be 5. In addition, b must be even, so the only possibilities are $b = 6, 4, 2$. Thus, there are 3 possible colonization options: $a = 3, b = 6$; $a = 4, b = 4$; $a = 5, b = 2$. In the first option, we take all 6 Mars-like planets, and can choose the Earth-like planets in $53 = 10$ ways. This gives us 10 possibilities. In the second option, we can choose any 4 of the 5 like Earth and any 4 of the 6 like Mars. This is $5464 = 75$ possibilities. In the third option, all Earth-like planets must be occupied while only 2 of those like Mars must be occupied. This is $5562 = 15$ possibilities. In all, there are $10 + 75 + 15 = 100$ planets.
- Note:* The answer correctly identifies the relationship between the number of colonies allocated to Earth-like and Mars-like planets, formulates the appropriate equation to

represent the total colonization units, and systematically derives the valid combinations of planet colonization that satisfy the constraints of the problem.

12. **Answer:** The exterior angles of a triangle sum to 360° , so the average of their measures is $360^\circ/3 = 120^\circ$.

Note: correct because the sum of all exterior angles of any polygon, including a triangle, is always 360 degrees, and dividing this total by the number of angles (three, in this case) gives the mean exterior angle, which is 120 degrees.